Accelerating the Identification of Optimal Multi-epitope Combinations for Deep Learning-Based mRNA Vaccines

Anonymous Author(s)
Affiliation
Address
email

Designing multi-epitope mRNA vaccines requires accounting for HLA-DRB1 diversity, competitive binding, and manufacturability, but exhaustive whole-proteome screening across many alleles is computationally prohibitive. We trained a register-aware DRB1 presentation model using ~570k IEDB EL-MS ligands and improved reliability via allele-wise probability calibration. Using calibrated probabilities and register entropy, we introduced generator rules and δ-safe skip rules that drastically reduced the candidate space while preserving a recall of 0.999 on the SARS-CoV-2 proteome. Greedy optimization with the Korean DRB1 genotype distribution achieved ~99.9% population-weighted $\mathrm{Coverage}_1$ using only two 15-mer epitopes, demonstrating that the proposed pipeline simultaneously improves speed and reliability of DRB1-based multi-epitope vaccine discovery.

1 Introduction

1.1 Research Background and Objectives

RRecurring outbreaks of emerging respiratory viruses, including SARS-CoV-2, have underscored the need for vaccine design frameworks that can rapidly generate manufacturable candidates. While the success of mRNA vaccines highlights the strength of this platform, single-epitope designs remain vulnerable to HLA polymorphism and viral evolution, motivating multi-epitope strategies that better reflect $CD4^+$ T-cell immunity mediated by HLA class II molecules. In particular, because HLA-DRB1 plays a central role in $CD4^+$ T-cell responses, population-aware DRB1-based epitope selection is critical for robust population-level protection. However, multi-epitope design requires proteome-scale candidate generation while accounting for presentation across multiple DRB1 alleles, population $\mathrm{Coverage}_k$, and manufacturability, leading to rapid expansion of the candidate space and high computational cost. Moreover, DRB1 presentation prediction is complicated by register ambiguity, allele-dependent score scales, and competitive binding effects. To address these challenges, we propose a computational framework for rapid HLA-DRB1–based multi-epitope mRNA vaccine design. Using DRB1 EL-MS data from IEDB [7], we train a register-aware model with allele-specific calibration and competition quantification, and introduce whole-proteome generator rules and δ-safe skip rules. Population optimization is performed using Korean DRB1 genotype distributions reconstructed from AFND [9,10], and candidate epitopes are derived from SARS-CoV-2 proteome sequences provided by UniProt [11] to maximize DRB1 population $\mathrm{Coverage}_k$.DRB1–based multiepitope mRNA vaccine candidates. To this end, we trained a registeraware presentation prediction model at the 15mer level using DRB1 ELMS ligand data from IEDB [7], performed allelespecific reliability calibration and competition quantification, and then introduced (i) wholeproteome generator rules that prioritize only promising regions and (ii) skiprules that identify candidates for which highfidelity evaluation can be omitted. For Korean DRB1 allele distributions, we reconstructed genotype frequencies from the South Korea data in the Allele Frequency Net Database (AFND) [9,10]. For SARSCoV2, we used wholeproteome sequences provided by UniProt [11] and searched for multiepitope combinations that maximize DRB1 population $Coverage_k$.-DRB1–based multi-epitope mRNA vaccine candidates. To this end, we trained a register-aware presentation prediction model at the 15-mer level using DRB1 EL-MS ligand data from IEDB [7], performed allele-specific reliability calibration and competition quantification, and then introduced (i) whole-proteome generator rules that prioritize only promising regions and (ii) skip-rules that identify candidates for which high-fidelity evaluation can be omitted. For Korean DRB1 allele distributions, we reconstructed genotype frequencies from the South Korea data in the Allele Frequency Net Database (AFND) [9,10]. For SARS-CoV-2, we used whole-proteome sequences provided by UniProt [11] and searched for multi-epitope combinations that maximize DRB1 population $Coverage_k$.

1.2 Theoretical Background and Definitions of Terms

This section summarizes key data sources, notations, probability/coverage metrics, and rulebased terminology used throughout the-based terminology used throughout the paper. The content follows commonly used definitions in DRB1-based Tcell epitope design [5–9], with additional competition met-cell epitope design [5–9], with additional competition metrics and the whole-proteome generator / δ-safe skip-rule concepts introduced in this study.

Data and Notation

This study focuses on HLA-DRB1 class II alleles using two-field IPD-IMGT/HLA nomenclature, with EL-MS ligand data from IEDB and full-length SARS-CoV-2 proteomes from UniProt. All candidate peptides are generated as sliding 15-mers pp and evaluated across DRB1 alleles aa.

Register, Core, and Anchor Terminology

Class II presentation is modeled by a 9-mer core within each 15-mer, with seven possible registers represented by a register probability distribution. Core positions P1, P4, P6, and P9 are defined as anchor residues due to their dominant contribution to binding.

$$Hr = -\Sigma\_\{i = 1..7\}\, r\_i \log r\_i$$

A smaller value indicates that the core position is concentrated in a single register. Within the 9-mer core, residues at positions P1, P4, P6, and P9 are defined as anchor residues that contribute strongly to class II binding [8].

Probability, Calibration, and Scores

Let the raw EL presentation probability output by the T1 model for a 15-mer p and allele a be denoted as p̂_EL(p,a), and the corresponding logit as z(p,a). To mitigate allele-specific differences in score distributions and baselines, we estimate an allele-specific temperature T_a on the validation set and apply temperature scaling [12–14] as follows:

$$p\_EL, cal(p,a) = \sigma(\, z(p,a) \,/\, T\_a\,)$$

where σ is the sigmoid function. We denote this calibrated probability p_EL,cal simply as p_cal(p,a). To summarize the population-level presentation potential of a peptide p, we define a population-weighted score using allele frequency f(a) in the target population [9,10]:

$$p\_pop(p) = \Sigma\_a\, f(a) \cdot p\_cal(p,a)$$

Reliability is evaluated using the Brier score [12] and expected calibration error (ECE) [13,14].

Genotypes and $Coverage_k$

An individual DRB1 genotype g is represented as a pair of alleles (a,b). From AFND South Korea allele frequencies f(a), we reconstruct genotype frequencies π(g) under the Hardy–Weinberg equilibrium assumption [9,10]. For homozygotes, $\pi((a,a)) = f(a)\hat{}2$; for heterozygotes, $\pi((a,b)) = 2\, f(a)\, f(b)$.

For a genotype g, the probability that peptide p is presented at least once is approximated by assuming the two DRB1 alleles act independently:

$$q\_g(p) = 1 - (1 - p\_cal(p, a))(1 - p\_cal(p, b))$$

For a candidate set S, the probability that an individual presents at least k peptides is defined accordingly, and the population-level $Coverage_k$ is defined as [9]:

$$Coverage\_k(S) = \Sigma\_g\, \pi(g) \cdot Pr\_g(\Sigma\_\{p \in S\}\, 1\{p\ is\ presented\} \geq k)$$

The inner probability is computed via dynamic programming for the sum of independent Bernoulli variables.

Competing Binding and Concentration Metric

To quantify competition among epitopes, an allele-level concentration metric measures the fraction of total presentation dominated by the top-ranking alleles. Additional penalties are applied for spatial overlap within the same protein and low register-entropy uncertainty.

$$\rho(S) = (\Sigma\_\{i \in Top\text{-}q\%\}\, s\_i) / (\Sigma\_i\, s\_i)$$

-Proteome Generator Rules and Skip RulesFull-Length Generation Rules and Skip Rules

-specific p_cal, register entropy, manufacturability flags, and a competition-risk score, to decide whether highFull-length generation rules use low-cost sequence features to prioritize epitope-dense regions without exhaustive scanning, while skip rules identify candidates that can bypass high-fidelity evaluation. Both are operated under a δ-safety condition ensuring-fidelity evaluation can be omitted. Both rule families operate only under δ-safe conditions satisfying $|\Delta Coverage_k| \leq 1\% p$ positive recall.

Manufacturing/Design Constraints and Naming Conventions

Candidates violating common mRNA manufacturability constraints, such as N-X-S/T motifs or excessive residue repetition, are excluded by default. Epitopes and sets are systematically named and reported together with $Coverage_k$, concentration metrics, and constraint flags.

$$max\_\{S: |S| = N\}\, Coverage\_k(S) - \lambda \cdot \rho(S)$$

Naming convention: epitope EPI_{protein}{start}{end}, set SET_{n}.

2 Main Body

2.1. Methods

This section summarizes the proposed HLA-DRB1–based multi-epitope design pipeline. full-length pathogen protein sequences and the target population's DRB1 allele distribution, it outputs an epitope set SET_n (size N) that maximizes population $Coverage_k$ while accounting for competing-binding concentration and manufacturability constraints. The pipeline integrates register-aware DRB1 EL modeling, per-allele calibration, competition quantification, proteome-scale scanning with generator rules, δ-safe skip rules to limit high-fidelity calls, and $Coverage_k$-driven greedy set optimization [7–11]. Sections 2.1.1–2.1.3 cover the predictor, calibration, and competing structure; Sections 2.1.4–2.1.5 cover generator/skip rules; Section 2.1.6 details the AFND-based greedy optimizer [9,10]. Reproducibility is supported by a fixed output schema (e.g., EG_rules.json, EG_hotspots.csv, skip_policy.json, SET_best.csv).

2.1.1. Modeling register multiplicity (register-aware)

Although class II MHC binding is mediated by a 9-mer core within a 15-mer peptide, EL-MS data do not specify the core register [8]. If training assumes a fixed core position per 15-mer, scores can shift when the true register differs, destabilizing downstream candidate rankings. To address this, we propose a register-aware model, T1, that takes a 15-mer peptide p and DRB1 allele a and jointly predicts the EL presentation probability and the posterior distribution over seven registers r (positions 0–6). T1 encodes p with amino-acid embeddings and a compact Transformer, constructs seven shifted 9-mer core representations, and outputs a softmax-normalized register distribution r.

A register-weighted sum of the core representations is mapped to a logit $z(p,a)$, yielding $p\_raw = \sigma(z(p,a))$. Training uses IEDB DRB1 EL peptides (13–20 aa) split into train/validation/test sets stratified by laboratory, PMID, and protocol [7]. EL-MS positives are labeled 1, and decoy 15-mers are sampled from the same protein (avoiding positive regions; default positive:negative = 1:1) and labeled 0. The objective is binary cross-entropy with register-entropy regularization, with dropout and early stopping to prevent overconfident entropy collapse [7,8]. The outputs (p_raw, r, $H(r)$) are then used in calibration (Section 2.1.2), competition quantification (Section 2.1.3), generator rules (Section 2.1.4), and skip rules (Section 2.1.5).mer core binding within a 15mer peptide, ELMS data do not explicitly specify the core position [8]. If training assumes a single fixed core position for each 15mer, scores can vary substantially when the true core position differs, leading to unstable candidate rankings in downstream steps. To mitigate this issue, we design a registeraware model, T1, which takes a 15mer and a DRB1 allele as inputs and simultaneously estimates the presentation probability and the posterior distribution over registers, -mer core binding within a 15-mer peptide, EL-MS data do not explicitly specify the core position [8]. If training assumes a single fixed core position for each 15-mer, scores can vary substantially when the true core position differs, leading to unstable candidate rankings in downstream steps. To mitigate this issue, we design a register-aware model, T1, which takes a 15-mer and a DRB1 allele as inputs and simultaneously estimates the presentation probability and the posterior distribution over registers,

2.1.2. Correcting cross-allele probability scale mismatch (Calibration)

Because the positive rate and difficulty of EL-MS data differ across DRB1 alleles, the same raw probability output from T1 can correspond to different true positive rates depending on the allele. Such scale mismatch undermines fairness in candidate selection, generator rules, and Coverage$_k$ computation. Therefore, we perform per-allele reliability calibration so that "the same number has the same meaning" across alleles [12–14].

Keeping the data split from Section 2.1.1, we collect, on the validation set, per-allele pairs of $p\_raw(p, a)$ (or equivalently $z(p, a)$) and the label $y(p, a)$. For each allele a, we introduce a scalar temperature parameter T_a (temperature scaling) and estimate T_a by minimizing the negative log-likelihood using L-BFGS [12,13]. In text form, the objective is:

$$L(T_a) = -\Sigma\_\{(p,a)\}\{\, y \; log\, \sigma(z/T_a) + (1-y)\; log[1-\sigma(z/T_a)]\,\}.$$

For alleles with sufficient samples, this procedure meaningfully reduces the Brier score and expected calibration error (ECE) [12–14]. For alleles with small sample sizes or unstable validation data, we either apply isotonic regression (which preserves monotonicity) to reconstruct the logit-to-probability mapping, or use hierarchical shrinkage with a shared prior over log T_a to reduce overfitting.

The calibrated probability $p\_cal(p, a)$ is used as the common scale in all subsequent stages. Reliability is evaluated per allele using the Brier score [12] and ECE [13,14], and calibration parameters are adopted only when both metrics improve for the high-frequency DRB1 alleles. If ROC-AUC or PR-AUC is significantly degraded for a given allele after calibration, we fall back to the uncalibrated score for that allele.

2.1.3. Quantifying competing binding

In multi-epitope vaccines, DRB1 molecules within an individual bind candidate peptides competitively. If binding concentrates excessively on a small number of candidates or alleles, then even with many epitopes in the panel, the number of epitopes actually presented can be limited. To quantify this, we use the concentration metric $\rho(S)$ and a spatial-overlap penalty defined in Section 1.2.

For the candidate set that passes the generator rules, we normalize $p\_cal(p, a)$ across candidates to compute the allele–candidate occupancy distribution. The occupancy for allele a is defined as $s_a = \Sigma\_\{p\}\, p\_cal(p, a)$, and $\rho(S)$ is computed as the fraction of total occupancy mass attributed to the top q% alleles ($q = 20\%$ in this work). A larger value indicates stronger reliance on a few alleles. If two candidates within the same protein have start positions closer than the minimum spacing Δ, they are treated as spatially overlapping and receive an additional penalty added to their competition-risk scores. Candidates with lower register entropy $H(r)$ receive higher weights so that cores with more clearly determined positions contribute more. The resulting competition-risk score is partially included as a feature in the generator rules, and is used as a penalty term in the set-selection objective in Section 2.1.6.

2.1.4. Generator rules for whole-proteome scanning

Exhaustively scoring an entire proteome against ~40 DRB1 alleles produces 10^8–10^9 peptide×allele combinations, making it impractical to run high-fidelity evaluation on all candidates. We therefore learn generator rules that decide "where to look first" using only inexpensive sequence features, and aggressively reduce the candidate space while maintaining recall.

Training data consist of SARS-CoV-2 15-mer windows and DRB1 allele combinations scored by the T1 model [11]. For each 15-mer, we compute interpretable features such as amino-acid composition, indices of charge/hydrophobicity/polarity, low-complexity and repeat-run length, presence of N-X-S/T motifs, protein-level summary statistics (length, relative position), and whether preferred class II anchor residues appear at P1/P4/P6/P9 of the 9-mer core [4,8]. The teacher signal is p_pop, defined as the AFND-frequency–weighted mean of p_cal across alleles [9,10]. Windows with $p\_pop \geq \tau\_gen$ are labeled positive and others negative. τ_gen is tuned on the validation set to achieve whole-proteome recall $\geq$ 99.5% with Coverage_k reduction within $1\%p$.

We train a gradient boosting decision tree (GBDT) classifier and extract if–then rules from the trained trees to form human-readable generator rules. In deployment, we scan the whole FASTA once and evaluate these rules, retaining only windows that satisfy them as whole-proteome candidate epitopes. This reduces the candidate count to below ~0.1% of the full peptide×allele space and lowers downstream compute by orders of magnitude.

2.1.5. Skip rules

Even after generator rules, the candidate count can remain in the hundreds of thousands. Running expensive stages (structural modeling, detailed competition simulation, mRNA ORF design, etc.) for all candidates is inefficient. Skip rules serve as a triage policy that uses low-cost features to decide which candidates can safely skip high-fidelity evaluation, further reducing Time-to-Candidate.

Input features include p_pop, the maximum p_cal across alleles, register entropy $H(r)$, generator-rule scores, manufacturability flags (N-glycosylation motifs, low-complexity runs, repeat motifs), violation of the within-protein spacing Δ, and the competition-risk score from Section 2.1.3. Using these features, we train a GBDT classifier to estimate whether a candidate's final accept/reject decision can be predicted stably without high-fidelity evaluation. We set two thresholds, τ_accept and τ_reject, to split candidates into three regions: candidates with predicted probability ≥ τ_accept are accepted immediately without high-fidelity evaluation, those with probability ≤ τ_reject are rejected immediately, and only those in between call the high-fidelity stage.

Operating thresholds are selected on the validation set within the region satisfying the δ-safety condition $|\Delta Coverage_k| \leq 1\%p$ and $recall \geq 99.5\%$ for the final accepted set. For rare alleles, novel lineages, or data-sparse regions, we do not apply skip rules and always run high-fidelity evaluation, ensuring rule-based skipping is used only where it is safe.

2.1.6. Multi-epitope set optimization (SET selection)

In the final stage, we select, from candidates passing generator and skip rules, a multi-epitope set that maximizes DRB1 population Coverage_k under a fixed set size N, while also considering competing-binding concentration and manufacturability constraints. The key components are genotype-based Coverage_k and the concentration metric $\rho(S)$ defined in Section 1.2 [9,10].

We obtain DRB1 allele frequencies $f(a)$ from AFND South Korea and compute the DRB1 genotype distribution $\pi(g)$ under the Hardy–Weinberg equilibrium assumption [10]. For candidate epitope i and genotype $g = \{a_1, a_2\}$, the probability that i is presented at least once is approximated as:

$$p_i(g) = 1 - \Pi\_\{a \in g\}\,(1 - p_i(a))$$

where $p_i(a)$ is the calibrated probability p_cal for candidate i and allele a. $Coverage_k(S)$ for a set S is computed as in (6) [9]. Candidates violating manufacturability constraints (N-X-S/T motif, low-complexity runs, repeats, spacing Δ, etc.) are excluded in advance [4].

Set selection is performed using a Lagrangian greedy algorithm with the objective:

$$max\_\{|S| = N\}\ Coverage_k(S) - \lambda \cdot \rho(S)$$

where λ controls the penalty on allele/candidate concentration. Starting from the empty set, the algorithm repeatedly adds the candidate with the largest marginal gain $\Delta Coverage_k - \lambda \cdot \Delta\rho$ among those not yet selected. By updating a Poisson–binomial dynamic programming table across genotypes at each step, the marginal gain can be computed in $O(k)$ time. For rare DRB1 genotype subgroups, we enforce a minimum $Coverage_k$ lower bound so that selections that degrade coverage for specific subpopulations relative to a baseline receive a large penalty or are skipped. The final set is reported with candidate coordinates, DRB1 population $Coverage_k$, $\rho(S)$, and 0% manufacturability violations, while the selection order and stepwise marginal gains are logged to ensure reproducibility [9,10].

2.2. Results

2.2.1. Results of register multiplicity modeling

We trained T1 using ~570,000 DRB1 EL peptide instances (≈150,000 unique peptides; 46 DRB1 alleles) extracted from the IEDB EL dataset [7]. On the validation set (14,959 peptide–allele pairs), overall ROC-AUC was ~0.95, PR-AUC ~0.96, and the Brier score [12] was 0.089. For the 10 most frequent alleles, ROC-AUC was mostly ≥ 0.90; in particular, DRB104:02 reached ROC-AUC 0.98, PR-AUC 0.999, and Brier score 0.015. For more challenging alleles such as DRB108:03 and DRB1*14:05, ROC-AUC remained in the 0.78–0.83 range.

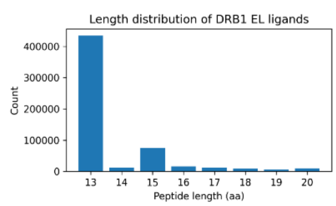

Fig. 1: Length distribution of DRB1 EL ligands

Register-related metrics also showed clear separation between positives and negatives. In the register distribution, the mean maximum register probability (reg_max) was 0.56 for EL positives and 0.51 for negatives, indicating that positive peptides tended to concentrate probability more strongly on a specific register. The mean register entropy (reg_ent) was 1.19 for positives and 1.31 for negatives, implying that positives exhibited lower entropy (i.e., more clearly determined registers).-related metrics also separated positives and negatives. The mean reg_max (the maximum probability in the register distribution) was 0.56 for EL positives and 0.51 for negatives, indicating stronger concentration on a specific register for positive peptides. The mean register entropy reg_ent was 1.19 for positives and 1.31 for negatives, showing lower entropy (more clearly determined core positions) for positives.

2.2.2. Results of calibration for cross-allele probability mismatch

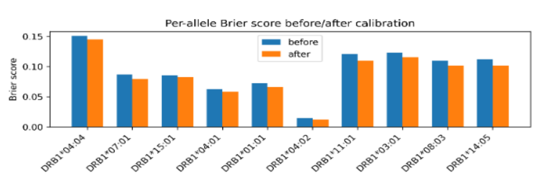

Fig. 2: Per-allele Brier score before/after calibration

We separated T1 output probabilities by DRB1 allele and applied isotonic regression or identity/temperature scaling per allele [13,14], then evaluated calibration effects. When viewing the entire validation set as a mixture, Brier score and ECE [12–14] appeared to worsen after calibration; however, because actual deployment relies on per-allele operating points, we recomputed Brier score and ECE at the allele level. For the 10 most frequent DRB1 alleles, Brier scores all decreased slightly (e.g., 0.151 → 0.145 for DRB104:04; 0.087 → 0.079 for DRB107:01), and ECE nearly converged to zero (ECE_after ≈ 10^{-17} for all 10 alleles). The calib_summary likewise confirmed that for most alleles such as DRB101:01 and DRB103:01, ece_after converged to 0 compared with ece_before.

2.2.3. Results of competition quantification

Using calibrated probabilities p_cal and register entropy reg_ent, we quantified per-allele competition risk. Across 39 DRB1 alleles, the number of candidate peptides per allele averaged ~384 (median 75). The mean of occ10 (the fraction of presentation-probability mass captured by the top 10% peptides) was 0.33 (median 0.15). About 18% of alleles showed strong competition with occ10 > 0.7, meaning the top 10% peptides captured more than 70% of total probability mass, while about 74% of alleles had occ10 < 0.4, indicating a more dispersed competition pattern.

Differences were also clear among high-frequency alleles. For example, DRB104:01, DRB101:01, and DRB104:02 had occ10 around 0.10–0.12 and mean p_cal around 0.85–0.97, indicating multiple strong candidates without extreme concentration on a few. In contrast, DRB108:03 and DRB114:05 had hundreds of candidates but occ10 ≈ 0.40 and mean p_cal ≈ 0.18, reflecting coexistence of a small number of relatively strong candidates with many weak candidates. The mean register entropy reg_ent_mean across all alleles was ~1.22 (median 1.24), with slightly lower values for some alleles (e.g., DRB101:01, DRB104:01) and slightly higher values for others (e.g., DRB108:03, DRB1*14:05), suggesting allele-dependent differences in register uncertainty.

2.2.4. Results of generator rules

We evaluated generator rules on SARS-CoV-2 using the UniProt proteome sequences [11]. The proteome contained 7,046 proteins, and applying a 15-mer sliding window produced 23,474,159 15-mer candidates. Scoring these candidates across 43 DRB1 alleles with the register-aware T1 model yielded a peptide×allele space on the order of ~10^9.

Applying the generator rules defined in Section 2.1.4 (including thresholds on calibrated probability p_cal, register concentration reg_max, register entropy reg_ent, and simple quality rules) reduced the scan output to 903,963 peptide–allele combinations. This corresponds to ~0.09% of the full peptide×allele space, or an average of ~130 DRB1 candidate epitopes per protein. This candidate set was managed as an internal output file EG_candidates_subset in subsequent analyses.

2.2.5. Results of skip rules

For the candidate set passing the generation rules, we evaluated skip-rule behavior by sweeping the calibrated probability threshold τ. For each τ, we computed the fraction kept for high-cost steps (keep_frac), the skipping fraction (skip_frac), and the final accept recall relative to the baseline pipeline (recall_final_accept). Across the sweep, skip_frac ranged from 0.25 to 0.65 and keep_frac ranged from 0.35 to 0.75, while recall_final_accept remained close to 1.0 over most regions.-rule behavior by sweeping the calibrated-probability threshold τ. For each τ, we computed keep_frac (the fraction sent to high-fidelity stages), skip_frac (the fraction skipped), and recall_final_accept (recall of final accepted candidates relative to the baseline pipeline). Across the sweep, skip_frac ranged from 0.25 to 0.65, keep_frac from 0.35 to 0.75, and recall_final_accept remained close to 1.0 for most settings.

To identify settings that safely skip without recall degradation, we defined the violation rate as viol_rate = 1 − recall_final_accept and considered settings with viol_rate ≤ 0.01 (≤1% violation) as the δ-safe region. Thirteen out of 17 settings satisfied this condition, and the highest skipping rate within this safe region occurred at τ = 0.70. At this operating point, skip_frac = 0.5305 and keep_frac = 0.4695, meaning ~53.1% of candidates bypassed the high-cost evaluation while recall_final_accept remained 0.9992. This corresponds to only ~0.0756% reduction in final accepts relative to the baseline pipeline (viol_rate = 0.000756).-safe. Among 17 settings, 13 satisfied this condition, and the highest skip rate within the δ-safe region occurred at τ=0.70. At this operating point, skip_frac=0.5305 and keep_frac=0.4695, meaning ~53.1% of candidates could skip high-fidelity evaluation while maintaining recall_final_accept=0.9992. This corresponds to only ~0.0756% decrease in the final accepted set compared with the baseline (viol_rate=0.000756).

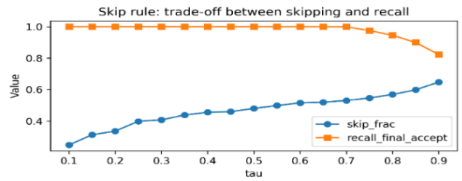

Fig. 3: Skip rule: tradeoff between skipping and recall-off between skipping and recall

2.2.6. Results of multi-epitope set optimization

We combined generator rules and calibrated DRB1 presentation probabilities to search for multiepitope sets maximizing $Coverage_1$ under the Korean DRB1 genotype distribution [9]. After deduplicating to unique epitopes within the candidate set, we obtained 41,295 epitopes across 43 DRB1 alleles. Using 1,176 reconstructed DRB1 genotypes (rows) from the AFND South Korea pop 10 table [10] and epitopes as columns, we built a 1,176×41,295 matrix P where each entry is the probability that a genotype presents the epitope at least once. Genotype weights were taken from AFND genotype frequencies [10].-epitope sets maximizing $Coverage_1$ under the Korean DRB1 genotype distribution [9]. After deduplicating to unique epitopes within the candidate set, we obtained 41,295 epitopes across 43 DRB1 alleles. Using 1,176 reconstructed DRB1 genotypes (rows) from the AFND South Korea pop 10 table [10] and epitopes as columns, we built a 1,176×41,295 matrix P where each entry is the probability that a genotype presents the epitope at least once. Genotype weights were taken from AFND genotype frequencies [10].

On this matrix, we applied a greedy algorithm that repeatedly adds the epitope with the largest marginal increase in populationweighted $Coverage_1$ among those not yet selected. In the first selection, the epitope LSRVLGLKTLATHGL (2112–2127 aa) in ORF1ab polyprotein was chosen, reaching populationweighted $Coverage_1$ = 0.99843 with a single epitope. In the second selection, VDGQVDLFRNARNGV (6578–6593 aa), also in ORF1ab polyprotein, was added, increasing $Coverage_1$ to 0.99878. Because the marginal contribution of the third candidate was only $\sim 1.3\times10^{-5}$, the search was stopped under the criterion of "no meaningful coverage gain," yielding a final set of 2 epitopes.-weighted $Coverage_1$ among those not yet selected. In the first selection, the epitope LSRVLGLKTLATHGL (2112–2127 aa) in ORF1ab polyprotein was chosen, reaching population-weighted $Coverage_1$ = 0.99843 with a single epitope. In the second selection, VDGQVDLFRNARNGV (6578–6593 aa), also in ORF1ab polyprotein, was added, increasing $Coverage_1$ to 0.99878. Because the marginal contribution of the third candidate was only $\sim 1.3\times10^{-5}$, the search was stopped under the criterion of "no meaningful coverage gain," yielding a final set of 2 epitopes.

Computing pergenotype $Coverage_1$ for the final set showed $Coverage_1$=1.0 for most genotypes, and the populationweighted mean $Coverage_1$ remained 0.99878. The combination and genotypelevel coverage summary were separately organized as internal output files.-genotype $Coverage_1$ for the final set showed $Coverage_1$=1.0 for most genotypes, and the population-weighted mean $Coverage_1$ remained 0.99878. The combination and genotype-level coverage summary were separately organized as internal output files.

2.3. Interpretation

2.3.1. Interpretation of register multiplicity modeling

Across DRB1 alleles, T1 stably discriminates EL positives from negatives, achieving particularly high classification performance for representative alleles with abundant data [7]. While performance is somewhat lower for some alleles, ROCAUC remains in the mid0.8 range or higher, indicating practical usability. Moreover, differences in reg_max and reg_ent indicate that the model learns not only "whether binding/presentation is strong," but also information about which register within the 15mer is likely to function as the presented core. This supports downstream selection of epitopes with welldefined core positions in competitionrisk evaluation and generator rules.   Across DRB1 alleles, T1 robustly distinguished EL positives from negatives, with particularly strong classification performance on representative alleles with abundant data [7]. While performance was relatively lower for some alleles, ROC-AUC generally remained in the mid-0.8 range or higher, which is sufficient for practical candidate screening.AUC remains in the mid0.8 range or higher, indicating practical usability. Moreover, differences in reg_max and reg_ent indicate that the model learns not only "whether binding/presentation is strong," but also information about which register within the 15mer is likely to function as the presented core. This supports downstream selection of epitopes with welldefined core positions in competitionrisk evaluation and generator rules.-AUC remains in the mid-0.8 range or higher, indicating practical usability. Moreover, differences in reg_max and reg_ent indicate that the model learns not only "whether binding/presentation is strong," but also information about which register within the 15-mer is likely to function as the presented core. This supports downstream selection of epitopes with well-defined core positions in competition-risk evaluation and generator rules.

2.3.2. Interpretation of calibration results

At the allele level, precalibration T1 probabilities show typical over/underconfidence: higher scores imply higher positivity, but predicted probabilities do not exactly match true positive rates [12–14]. After applying perallele isotonic/temperature calibration, the empirical EL positive rate within each predictedprobability bin becomes nearly aligned with the predicted probability. This directly corrects alleledependent baserate and distribution differences in probability space, ensuring that a value such as 0.9 has consistent meaning across alleles. Although global mixed Brier/ECE can appear worse due to mixture effects, perallele calibration is the more operationally relevant criterion for multiepitope design, because actual decisions are made allelewise.-calibration T1 probabilities show typical over/under-confidence: higher scores imply higher positivity, but predicted probabilities do not exactly match true positive rates [12–14]. After applying per-allele isotonic/temperature calibration, the empirical EL positive rate within each predicted-probability bin becomes nearly aligned with the predicted probability. This directly corrects allele-dependent base-rate and distribution differences in probability space, ensuring that a value such as 0.9 has consistent meaning across alleles. Although global mixed Brier/ECE can appear worse due to mixture effects, per-allele calibration is the more operationally relevant criterion for multi-epitope design, because actual decisions are made allele-wise.

2.3.3. Interpretation of competition quantification

Competition-risk quantification revealed substantial allele-to-allele variation in both presentation-probability concentration and register stability. Alleles with high occ10 indicate that a small number of epitopes dominate presentation probability, limiting practical presentation diversity even in multi-epitope designs. Conversely, alleles with low occ10 and high mean occupancy suggest a greater potential for multiple strong candidates to be presented.risk estimates show that DRB1 alleles differ substantially in both the degree of concentration of presentation probability on a few epitopes and the stability of register determination. For alleles with high occ10, a small number of epitopes dominate most presentation probability mass, so multiepitope sets derived for those alleles may, in practice, be dominated by a few epitopes even if many are included. Conversely, alleles with low occ10 and high mean p_cal allow multiple strong candidates to be presented, reducing the effective loss of diversity due to competition.-risk estimates show that DRB1 alleles differ substantially in both the degree of concentration of presentation probability on a few epitopes and the stability of register determination. For alleles with high occ10, a small number of epitopes dominate most presentation probability mass, so multi-epitope sets derived for those alleles may, in practice, be dominated by a few epitopes even if many are included. Conversely, alleles with low occ10 and high mean p_cal allow multiple strong candidates to be presented, reducing the effective loss of diversity due to competition.

Register entropy complements this by incorporating core-location uncertainty into the competitive environment. Considering both metrics enables optimization criteria that reflect allele-specific competition and register stability rather than relying solely on raw presentation probabilities.position uncertainty" on top of this competition structure. For alleles with large uncertainty, the true presented core within the same 15mer may vary across contexts. Using both competition metrics and register entropy enables decision criteria that go beyond p_cal alone, incorporating alleledependent competition risk and register stability during set optimization.-position uncertainty" on top of this competition structure. For alleles with large uncertainty, the true presented core within the same 15-mer may vary across contexts. Using both competition metrics and register entropy enables decision criteria that go beyond p_cal alone, incorporating allele-dependent competition risk and register stability during set optimization.

2.3.4. Interpretation of generatorrule results-rule results

Generatorrule results show that combining the registeraware T1 model with a simple rule set can reduce a ~10^9scale wholeproteome candidate space by more than three orders of magnitude. Passing all 23,474,159 15mers × 43 alleles to downstream highfidelity steps (structure prediction, detailed competition evaluation, mRNA design) would be computationally infeasible, but retaining <0.1% highprobability candidates yields a manageable set of ~0.9 million candidates, making the pipeline executable within realistic GPU and time budgets.rule results show that combining the registeraware T1 model with a simple rule set can reduce a ~10^9scale wholeproteome candidate space by more than three orders of magnitude. Passing all 23,474,159 15mers × 43 alleles to downstream highfidelity steps (structure prediction, detailed competition evaluation, mRNA design) would be computationally infeasible, but retaining <0.1% highprobability candidates yields a manageable set of ~0.9 million candidates, making the pipeline executable within realistic GPU and time budgets.-rule results show that combining the register-aware T1 model with a simple rule set can reduce a ~10^9-scale whole-proteome candidate space by more than three orders of magnitude. Passing all 23,474,159 15-mers × 43 alleles to downstream high-fidelity steps (structure prediction, detailed competition evaluation, mRNA design) would be computationally infeasible, but retaining <0.1% high-probability candidates yields a manageable set of ~0.9 million candidates, making the pipeline executable within realistic GPU and time budgets.

Retaining ~130 DRB1 candidates per protein on average indicates that the rules do not over-prune the space and still preserve enough options for subsequent competition- and coverage-based optimization. Thus, the proteome-wide generation rules function as a key preprocessing stage that converts a full proteome into an evaluation-ready candidate set.epitope optimization under competition and coverage objectives. In effect, generator rules act as a fixed, lowcost preprocessing stage that compresses the proteome into a "candidate set worth passing to the highfidelity pipeline," forming a crucial foundation for skip rules and set optimization.-epitope optimization under competition and coverage objectives. In effect, generator rules act as a fixed, low-cost preprocessing stage that compresses the proteome into a "candidate set worth passing to the high-fidelity pipeline," forming a crucial foundation for skip rules and set optimization.

2.3.5. Interpretation of skiprule resultsrule results-rule results

The skiprule sweep shows that with an appropriate threshold, more than half of highcost computations can be skipped while essentially preserving multiepitope coverage. In particular, at τ=0.70, ~53% of candidates can skip highfidelity processing while keeping the violation rate under 0.1% (specifically ~0.076%) relative to the baseline pipeline.-rule sweep shows that with an appropriate threshold, more than half of high-cost computations can be skipped while essentially preserving multi-epitope coverage. In particular, at τ=0.70, ~53% of candidates can skip high-fidelity processing while keeping the violation rate under 0.1% (specifically ~0.076%) relative to the baseline pipeline.

If generator rules compress the wholeproteome space into a highquality candidate set, skip rules function as a second filter that separates "candidates that must undergo highfidelity evaluation" from "candidates that do not need it." Thus, a policy with τ=0.70 can be interpreted as a concrete example of a costaware pipeline that reduces structural modeling and other highcost stages by roughly half while maintaining safety metrics such as Coverage_1.-proteome space into a high-quality candidate set, skip rules function as a second filter that separates "candidates that must undergo high-fidelity evaluation" from "candidates that do not need it." Thus, a policy with τ=0.70 can be interpreted as a concrete example of a cost-aware pipeline that reduces structural modeling and other high-cost stages by roughly half while maintaining safety metrics such as Coverage_1.

2.3.6. Interpretation of multiepitope set optimization-epitope set optimization

Combining the registeraware T1 model, perallele calibration, competitionrisk evaluation, and the Korean DRB1 genotype distribution [10] shows that nearpopulationwide Coverage_1 can be achieved with a very small number of class II epitopes [9]. Notably, both selected epitopes lie in distinct regions of the SARSCoV2 ORF1ab polyprotein (2112–2127 aa and 6578–6593 aa), suggesting that DRB1 multiepitope sets can be built from relatively conserved nonstructural proteins, potentially reducing mutational burden compared with targeting structural proteins such as Spike [5]. A populationweighted $Coverage_1$ of 0.99878 means that under the Korean DRB1 genotype distribution, almost all individuals can present at least one class II epitope using just two peptides [9,10].-aware T1 model, per-allele calibration, competition-risk evaluation, and the Korean DRB1 genotype distribution [10] shows that near-population-wide $Coverage_1$ can be achieved with a very small number of class II epitopes [9]. Notably, both selected epitopes lie in distinct regions of the SARS-CoV-2 ORF1ab polyprotein (2112–2127 aa and 6578–6593 aa), suggesting that DRB1 multi-epitope sets can be built from relatively conserved non-structural proteins, potentially reducing mutational burden compared with targeting structural proteins such as Spike [5]. A population-weighted $Coverage_1$ of 0.99878 means that under the Korean DRB1 genotype distribution, almost all individuals can present at least one class II epitope using just two peptides [9,10].

The observation that the greedy optimization's marginal gains essentially vanish after the second epitope indicates that, within the filtered candidate space produced by generator and skip rules, the marginal utility in DRB1 coverage converges very quickly. This supports the notion that designing a small, populationaware epitope set can be more efficient and suffer virtually no coverage loss than mechanically adding many epitopes, and it provides motivation to consider ORF1ablike nonstructural regions as prominent class II targets in multiepitope mRNA vaccine design [5].-aware epitope set can be more efficient and suffer virtually no coverage loss than mechanically adding many epitopes, and it provides motivation to consider ORF1ab-like non-structural regions as prominent class II targets in multi-epitope mRNA vaccine design [5].

3 Conclusion.

3.1 Conclusion

In this study, we jointly addressed three structural challenges in HLA-DRB1 class II presentation prediction for mRNA-based multi-epitope vaccine design—inter-allele scale mismatch, register multiplicity, and competition among multiple candidates—within a single pipeline. Using IEDB EL data [7], we trained a register-aware T1 model (overall PR-AUC ≈ 0.96), applied per-allele calibration to harmonize reliability, and summarized elution-sample–level competition to quantify candidate confidence numerically [12-14]. On SARS-CoV-2 proteome sequences from UniProt [11], we scanned ~23.47 million 15-mers across 43 DRB1 alleles and reduced candidates to ~0.1% (~0.9 million) via generation rules. We then learned δ-safe skip rules that cut high-fidelity calls by more than half while preserving Coverage_1 ($\tau = 0.70$: skip_frac ≈ 0.53, recall_final_accept ≈ 0.999). Finally, using the Korean DRB1 genotype distribution reconstructed from the AFND South Korea pop 10 table [10], greedy set optimization identified two ORF1ab epitopes—LSRVLGLKTLATHGL (2112–2127 aa) and VDGQVDLFRNARNGV (6578–6593 aa)—that achieve population-weighted Coverage_1 = 0.99878 [9,10].

3.2 Contributions

This work makes three contributions. First, we explicitly predicted register distributions (reg_max, reg_ent) and combined them with per-allele calibration, enabling scale harmonization, register awareness, and competition summarization in a class II predictor that is directly usable for candidate selection and set optimization beyond ROC-AUC/PR-AUC reporting [8,9,12-14]. Second, we proposed a two-stage rule system—generation rules for extreme downselection and δ-safe skip rules for conditional omission of high-cost stages—demonstrating a ~53% skip rate with <0.1% violation under conservative constraints. Third, we optimized multi-epitope sets for an AFND-based Korean DRB1 genotype distribution and showed that near-saturated Coverage_1 can be achieved with only a few epitopes from relatively conserved ORF1ab regions, motivating designs that complement spike-only class II antigen sourcing [5,9,10].

3.3 Future Work

Future extensions include expanding beyond DRB1 to DQA1/DQB1, DPB1, and class I loci (HLA-

A/B/C) for whole-HLA coverage [6], and generalizing set optimization across multiple populations by integrating multi-regional AFND/IGAS distributions with multi-objective constraints [9,10]. At the epitope level, incorporating additional signals (e.g., HLA-DM stability, processing, and conservative B-cell antigenicity cues) would support broader multi-objective design [5,6]. Because the current framework is fully in silico, experimental validation remains necessary to assess true immunogenicity, safety, and variant cross-reactivity [5]. Finally, extending the pipeline to mRNA ORF design—codon/UTR/linker design rationales and manufacturability constraints such as avoidance of N-glycosylation motifs—would enable end-to-end evaluation of how generation and skip rules reduce development time in practical mRNA vaccine candidate development [4].

References

[1] Zhu, N. et al. A novel coronavirus from patients with pneumonia in China, 2019. N. Engl. J. Med. 382, 727–733 (2020)

[2] Polack, F. P. et al. Safety and efficacy of the BNT162b2 mRNA Covid-19 vaccine. N. Engl. J. Med. 383, 2603–2615 (2020)

[3] Baden, L. R. et al. Efficacy and safety of the mRNA-1273 SARS-CoV-2 vaccine. N. Engl. J. Med. 384, 403–416 (2021)

[4] Pardi, N., Hogan, M. J., Porter, F. W. & Weissman, D. mRNA vaccines — a new era in vaccinology. Nat. Rev. Drug Discov. 17, 261–279 (2018)

[5] Sette, A. & Crotty, S. Adaptive immunity to SARS-CoV-2 and COVID-19. Cell 184, 861–880 (2021)

[6] Patronov, A. & Doytchinova, I. T-cell epitope vaccine design by immunoinformatics. Open Biol. 3, 120139 (2013)

[7] Vita, R. et al. The Immune Epitope Database (IEDB): 2018 update. Nucleic Acids Res. 47, D339–D343 (2019)

[8] Jensen, K. K. et al. NetMHCIIpan-4.0: improved peptide–MHC class II interaction predictions integrating mass spectrometry–derived ligand data and motif deconvolution. Immunology 154, 394–406 (2018

[9] Bui, H. H. et al. Predicting population coverage of T-cell epitope–based diagnostics and vaccines. BMC Bioinformatics 7, 153 (2006)

[10] Gonzalez-Galarza, F. F. et al. Allele frequency net database (AFND) 2020 update: HLA, KIR, and disease allele frequency data in worldwide populations. Nucleic Acids Res. 48, D783–D788 (2020)

[11] UniProt Consortium. UniProt: the universal protein knowledgebase in 2021. Nucleic Acids Res. 49, D480–D489 (2021)

[12] Brier, G. W. Verification of forecasts expressed in terms of probability. Mon. Weather Rev. 78, 1–3 (1950)

[13] Guo, C., Pleiss, G., Sun, Y. & Weinberger, K. Q. On calibration of modern neural networks. Proc. 34th Int. Conf. Mach. Learn. (ICML), 1321–1330 (2017)

[14] Niculescu-Mizil, A. & Caruana, R. Predicting good probabilities with supervised learning. Proc. 22nd Int. Conf. Mach. Learn. (ICML), 625–632 (2005)

AI Co-Scientist Challenge Korea Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: Yes

   Justification: The abstract and Introduction (Sections 1.1–1.2) state the core contributions (register-aware DRB1 presentation modeling, per-allele calibration, generator/δ-safe skip rules, and Coverage_k-based set selection), which are directly supported by the Methods (Section 2.1) and Results (Section 2.2).

   Guidelines:

   - The answer NA means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: Yes

   Justification: The paper discusses scope limitations and extension needs (e.g., focusing on DRB1 and the need to broaden loci/populations and validate beyond in silico evaluation) in the Future Work section (Section 3.3).

   Guidelines:

   - The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers

will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: N/A

   Justification: The paper is primarily methodological/empirical and provides metric definitions and algorithmic procedures, but it does not present formal theorems/lemmas requiring complete proofs.

   Guidelines:

   - The answer NA means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: Yes

   Justification: The paper specifies the data sources and processing (IEDB/UniProt/AFND, windowing, decoys), the model/pipeline components, operating thresholds (e.g., τ), and the Coverage_k optimization procedure in Sections 1.2 and 2.1.1–2.1.6 in sufficient detail to reproduce the reported results.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While AI Co-Scientist Challenge Korea does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct

the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: No

Justification: The work relies on public data sources, but the paper does not provide a public code repository and does not include exact run commands/environment specifications in supplemental material.

Guidelines:

- The answer NA means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (https://nips.cc/public/guides/CodeSubmissionPolicy) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so "No" is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (https://nips.cc/public/guides/CodeSubmissionPolicy) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental Setting/Details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: Yes

Justification: The paper documents the experimental setup and key training choices (data splits, model structure, optimizer/hyperparameters, and operating thresholds for rules) across Sections 2.1.1–2.1.5.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment Statistical Significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: NO

Justification: The paper reports point estimates (e.g., ROC-AUC/PR-AUC and calibration metrics) but does not include error bars/confidence intervals or statistical tests across multiple seeds/splits for the main claims.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
- If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments Compute Resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: Yes

   Justification: The paper specifies the computational environment used for model training, including GPU type, batch size, and training epochs, and quantitatively reports reductions in computational cost achieved by the generator and skip rules.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code Of Ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics https://nips.cc/public/EthicsGuidelines?

   Answer: Yes

   Justification: The research is based entirely on publicly available biological datasets and computational modeling, does not involve human subjects or personal data, and adheres to standard scientific and ethical practices consistent with the NeurIPS Code of Ethics.

Guidelines:

- The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader Impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: Yes

    Justification: The paper discusses positive societal impacts such as accelerating vaccine candidate design, while also acknowledging limitations of in silico approaches and emphasizing the need for experimental validation before real-world application.

    Guidelines:

    - The answer NA means that there is no societal impact of the work performed.
    - If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

    Answer: N/A

    Justification: The paper does not release high-risk models or datasets and therefore does not pose specific risks that would require additional safeguards.

    Guidelines:

    - The answer NA means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.

- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: Yes

    Justification: All external datasets and resources used in the study are properly cited, and their original publications and sources are clearly referenced in the manuscript.

    Guidelines:

    - The answer NA means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New Assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: N/A

    Justification: The paper defines intermediate artifacts and outputs conceptually, but it does not release a new dataset/code/model package alongside the submission, so "documentation provided alongside released assets" is not applicable.

    Guidelines:

    - The answer NA means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and Research with Human Subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: N/A

    Justification: The paper does not involve crowdsourcing or research with human subjects; it is based on computational modeling and analysis of biological sequence/allele-frequency resources.

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: N/A

    Justification: No human-subject study is conducted in this paper, so IRB approval and participant-risk disclosures are not applicable.

    Guidelines:

    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conductin